

Design Document: Developer & Robotics Portfolio Landing Page

Design Reference: [Dribbble UI/UX Concept](#)

Profile Target: Pyla Amruth Varun — Mechanical Robotics Engineer

1. Visual & Aesthetic Framework

- **Theme:** High-tech Developer/Robotics aesthetic utilizing dark-mode surfaces, subtle neon border accents, terminal-style metadata, and modular glassmorphic cards.
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- **Color Palette:**
- - Canvas Background: Deep Obsidian (`#0A0D14`)
 -
 - Card Surfaces: Translucent Slate (`#121722`) with subtle border strokes (`rgba(255, 255, 255, 0.08)`)
 -
 - Primary Accent: Electric Cyan (`#00F2FE`) & Neon Indigo (`#6366F1`)
 -
 - Text: High-contrast Off-White (`#F8FAFC`) and Muted Gray (`#94A3B8`)
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- **Typography:**
- - Headings: *Syne / Cabinet Grotesk*
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 - Body & Descriptions: *Inter / Plus Jakarta Sans*
 -
 - Code & Technical Tags: *JetBrains Mono*
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2. Information Architecture & Section Breakdown

3. Section Specifications

3.1 Sticky Navigation Bar

- Left: `varun://robotics` (Monospace branding).
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- Center: Nav links (`// Stack`, `// Projects`, `// Experience`, `// Contact`).
-
- Right:
- - `[Download CV]` (Direct download trigger for resume)
 -
 - `[Get in Touch]` (Electric Cyan accent button)
 -

3.2 Hero Section

- Left Column:
- - Status Pill: `● AVAILABLE FOR ROBOTICS & AUTOMATION ROLES`
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 - Main Headline:
 - - **ENGINEERING HARDWARE, KINEMATICS & AUTOMATION SYSTEMS**
 -
 - Sub-text:
 - - *"Mechanical Robotics graduate specialized in CATIA V5, SolidWorks, motion simulation, and rapid prototyping for industrial automation and smart hardware."*
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 - Actions:
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- [View Projects](#) ↓ (Filled cyan button)
-
- [LinkedIn Profile](#) ↗ (Ghost button linking to [\[linkedin.com/in/pyla-amruth-varun-33b938251\]\(https://linkedin.com/in/pyla-amruth-varun-33b938251\)](https://linkedin.com/in/pyla-amruth-varun-33b938251))
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- **Right Column (Interactive Terminal Card):**
-

```
Bash
    "Pyla Amruth Varun"
    "Mechanical Robotics" "EV Enclosures" "Kinematics"
    "Design Engineer @ DJT EV POWERTECH"
```

3.3 Technical Stack Bento Matrix

Categorized tool chips formatted in developer-style pills with icons:

- **3D CAD & Mechanical Design:** CATIA V5, SolidWorks, AutoCAD
-
- **Robotics & Simulation:** RoboAnalyzer, MATLAB Simulink, Blender Motion Animation
-
- **Manufacturing & Prototyping:** CNC Programming, MasterCAM, 3D Printing (Additive Manufacturing)
-
- **Software & Embedded:** Python, Betaflight Controller
-

3.4 Featured Projects (Developer Card Layout)

1. **3DOF RPR Robot Kinematic Analysis**
- 2.
3.
 - **Category:** *Robotics Simulation & Workspace Validation*
 -
 -
 - **Stack:** [MATLAB Simulink](#) • [Kinematics](#) • [RoboAnalyzer](#)
 -
 -
 - **Details:** Forward and inverse kinematic analysis to map robot motion behavior and workspace reachability.
 -
4. **EV Charging Station Enclosures**

5.

6.

- **Category:** *Industrial Design & DFM*
-
-
- **Stack:** CATIA V5 • AutoCAD • Sheet Metal
-
-
- **Details:** Enclosure design simplifying assembly processes and optimizing component integration for EV charging machines.
-

7. **Collaborative Robot (Cobot) Structure**

8.

9.

- **Category:** *Modular Robotics*
-
-
- **Stack:** CATIA V5 • Kinematic Analysis
-
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- **Details:** Designed modular joints and verified reach envelope for safe human-robot interaction.
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10. **Articulating Laparoscopic Surgical Tool**

11.

12.

- **Category:** *Biomedical Product Design*
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-
- **Stack:** CATIA V5 • 3D Printing Prototyping
-
-
- **Details:** Multi-axis joint mechanism for minimally invasive surgical instruments with ergonomic handle optimization.
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13. Custom Drone with Betaflight Avionics

14.

15.

- **Category:** *Aerial Robotics*
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-
- **Stack:** Betaflight • Avionics Integration • CAD
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-
- **Details:** Structural assembly and controller tuning for enhanced flight stability.
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3.5 Experience Timeline

- **DJT EV POWERTECH** — *Design Engineer (05/2025 – Present)*
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- - Designed EV charging machine enclosures using CATIA V5.
 -
 - Simulated automation machines and Montratec systems for QFN packaging lines.
 -
 - Produced site planning layouts in AutoCAD for equipment deployment.
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- **Hunumax Technologies** — *Intern (12/2024 – 03/2025)*
-
- - Modeled articulating laparoscopic instruments with enhanced mobility in CATIA V5.
 -
 - Optimized CAD models for additive manufacturing and physical fit verification.
 -
- **Dreambots** — *Intern (04/2024 – 05/2024)*
-
- - Designed cobot arm assemblies and executed kinematic workspace reach analysis.
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3.6 Education & Key Highlights Bento

- **Education:** B.Tech in Mechanical Robotics, Gayatri Vidya Parishad College of Engineering (11/2021 – 05/2025)
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- **Certifications:**
- - CNC Training Program (APSSDC & Siemens)
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 - Python Programming Course
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- **Leadership:** Basketball Captain — MRB Basketball Team (Annual Cup)
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3.7 Terminal Footer & Direct Contact

- **Direct Email:** amruthvarunpyla@gmail.com (with 1-click copy action)
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- **Phone:** [+91 9052926058](tel:+919052926058)
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- **Location:** Visakhapatnam, Andhra Pradesh
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- **CLI Quick Commands:** Interactive mock terminal buttons (`$ ping amruth`, `$ cat contact.json`, `$ open linkedin`).
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4. Tech Stack Recommendation

- **Framework:** Next.js 15 (App Router) + React
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- **Styling:** Tailwind CSS + Lucide Icons
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- **Animation:** Framer Motion (for terminal typing effect and card hover glows)
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- **Hosting:** Vercel